Sri Lanka Institute of Information Technology

IT2130 – Operating Systems and System Administration
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Practical Answer Submission

Practical Sheet No: 05

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Activity 1

Consider the below program and answer the questions.

```
  GNU nano 7.2                                          b1.c *
#include <stdio.h>
#include <stdio.h>

void* print_message(void* arg){
 printf("Hello from thr thread!\n");
 pthread_exit(NULL);
}

int main(){
 pthread_t thread_id;

 if(pthread_create (&thread_id, NULL, print_message, NULL) !=0){
  perror("Thread creation failed");
  return 1;
 }

 pthread_join(thread_id, NULL);

 printf("Thread has finished execution.\n");
 return 0;

}
```

a. What is the output you can see in the terminal?

```
kamshiga@kamshiga-VirtualBox:~$ nano b1.c
kamshiga@kamshiga-VirtualBox:~$ gcc -o b1 b1.c -lpthread
kamshiga@kamshiga-VirtualBox:~$ ./b1
Hello from the thread!
Thread has finished execution.
kamshiga@kamshiga-VirtualBox:~$
```

b. What are the functions you found in the above program?

- pthread_create() - creates a new thread
- pthread_join() - waits for a thread to finish
- pthread_exit() - terminates a thread
- printf() - prints output
- main() - program entry point

c. Briefly describe the above program

- This program demonstrates basic thread creation using POSIX threads (pthreads).
  The main thread creates a child thread, both threads execute independently, print messages, and then terminate after synchronization using pthread_join().

Activity 2

Write a C program to create 3 threads. Each thread should print its thread number.

```
  GNU nano 7.2                                                    b2.c
#include <stdio.h>
#include <pthread.h>

void*  threadFunc1(void* arg){
 printf("Thread 1 is running\n");
 return NULL;
}
void* threadFunc2(void* arg){
 printf("Thread 2 is running\n");
 return NULL;
}
void* threadFunc3(void* arg){
 printf("Thread 3 is running\n");
 return NULL;
}

int main(){
 pthread_t t1, t2, t3;

 pthread_create(&t1, NULL, threadFunc1, NULL);
 pthread_create(&t2, NULL, threadFunc2, NULL);
 pthread_create(&t3, NULL, threadFunc3, NULL);

 pthread_join(t1, NULL);
 pthread_join(t2, NULL);
 pthread_join(t3, NULL);
 return 0;
}
                                                  [ Wrote 28 lines ]
```

```
kamshiga@kamshiga-VirtualBox: ~
kamshiga@kamshiga-VirtualBox:~$ nano b2.c
kamshiga@kamshiga-VirtualBox:~$ gcc -o b2 b2.c
kamshiga@kamshiga-VirtualBox:~$ ./b2
Thread 1 is running
Thread 2 is running
Thread 3 is running
kamshiga@kamshiga-VirtualBox:~$
```

Activity 3

Consider the program given below and answer the questions.

```
  GNU nano 7.2                                          b3.c
#include <stdio.h>
#include <pthread.h>

typedef struct {
 int a;
 int b;
} Numbers;

void* calculate(void* arg){
 Numbers* nums= (Numbers*)arg;
 int sum = nums->a +nums->b;
 printf("Sum = %d\n", sum);
 return NULL;
}

int main() {
 pthread_t t;

 Numbers n;
 n.a = 10;
 n.b = 20;

 pthread_create(&t, NULL, calculate, &n);
 pthread_join(t, NULL);

 return 0;
                                                [ Read 28 lines ]
```

```
kamshiga@kamshiga-VirtualBox:~$ nano b3.c
kamshiga@kamshiga-VirtualBox:~$ gcc -o b3 b3.c
kamshiga@kamshiga-VirtualBox:~$ ./b3
Sum = 30
kamshiga@kamshiga-VirtualBox:~$
```

a. What is the purpose of the typedef struct in this program?
   - typedef struct is used to group multiple related variables(in here like a, b) into a single structure so they can be passed easily to a thread.

b. Why must the thread function parameter be of type void*?
   - POSIX threads require the thread function to accept any type of data.
   - void* allows passing integers, structures, arrays, etc.

c. Explain the following line: Numbers* nums = (Numbers*)arg; lines:
   - performs “type casting”. It converts the generic ‘void* arg’ back into a specific pointer of type ‘Numbers*’ so that the program can correctly access the struct members (‘nums-a’ and ‘nums-b’) within the thread function.

Activity 4

Consider the program in activity 3 and modify it according to the questions asked below.

a. Get the values for a and b as keyboard inputs.

b. Implement three threads for the below tasks.

i. Create the first thread to calculate the sum of the values.

ii. Create the second thread to calculate the product of the values.

iii. Pass an array of 4 numbers and assign the third thread to calculate the sum of them.

```
GNU nano 7.2                                                    b4.c *
#include <stdio.h>
#include <pthread.h>

typedef struct{
 int a;
 int b;
}Numbers;

typedef struct{
 int arr[4];
}ArrayData;

Numbers nums;
ArrayData data;

void* sumThread(void* arg){
 printf("Sum= %d\n",nums.a + nums.b);
 return NULL;
}

void* productThread(void* arg){
 printf("Product= %d\n",nums.a * nums.b);
 return NULL;
}

void* arraySumThread(void* arg){
 int sum=0;
 for (int i=0;i<4;i++){
  sum+= data.arr[i];
 }
 printf("Array Total Sum= %d\n",sum);
 return NULL;
}
```

```
int main(){
 pthread_t t1,t2,t3;

 printf("Enter two numbers: ");
 scanf("%d %d",&nums.a, &nums.b);

 printf("Enter 4 numbers for the array: ");
 for(int i=0; i<4; i++){
  scanf("%d",&data.arr[i]);
 }

 pthread_create(&t1, NULL, sumThread, NULL);
 pthread_create(&t2, NULL, productThread, NULL);
 pthread_create(&t3, NULL, arraySumThread, NULL);

 pthread_join(t1, NULL);
 pthread_join(t2, NULL);
 pthread_join(t3, NULL);

 return 0;
}
```

```
kamshiga@kamshiga-VirtualBox:~$ nano b4.c
kamshiga@kamshiga-VirtualBox:~$ gcc -o b4 b4.c
kamshiga@kamshiga-VirtualBox:~$ ./b4
Enter two numbers: 5 4
Enter 4 numbers for the array: 1 2 3 4
Sum= 9
Product= 20
Array Total Sum= 10
kamshiga@kamshiga-VirtualBox:~$
```